5th Oct 2022 Wednesday

TRANSISTOR

- BJT (Bipolar Junction Transistor)
  - NPN
  - PNP
- MOSFET
  - P-channel
  - N-channel

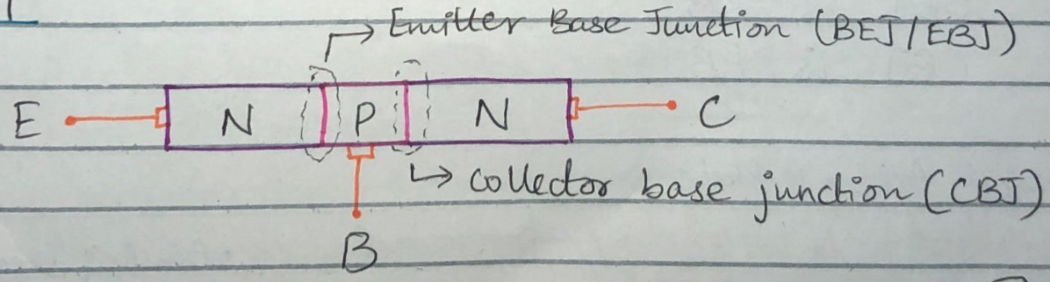
→ Three terminal non-linear device

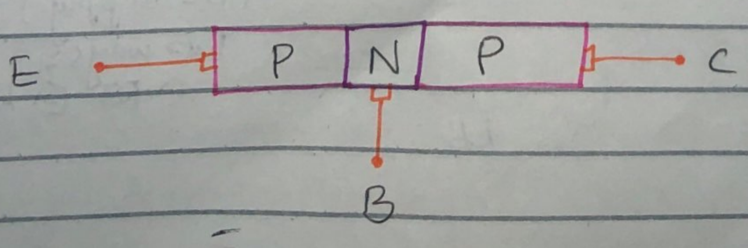
* BJT → Emitter, Base, Collector
* MOSFET → Source, Gate, Drain

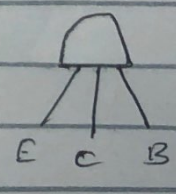
Applications

- amplifier
- Switch

* BJT







Modes of Operation : (depend on biasing).

① **Forward Active / Amplication Mode** (Amplifier)
↓
BJT used as

EBJ / BEJ is FB
CBJ is RB

② **Saturation Mode** (ON switch)

Both BEJ and CBJ are FB.

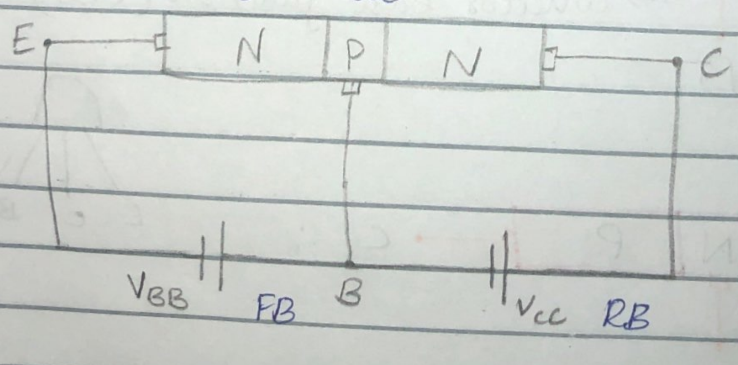
③ **Cut off Mode** (OFF switch)

Both EBJ and CBJ are RB

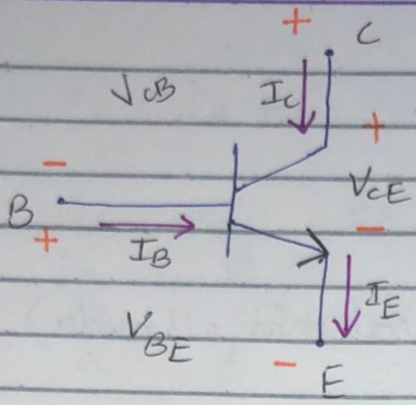
④ **Reverse Active / Inverse Mode**

BEJ is RB
CBJ is FB

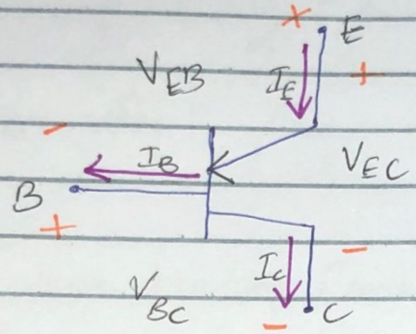
Forward Active Mode



• Biasing means connecting a source (DC supply)
Two ways:
① FB ② RB



NPN



PNP

☆ $\boxed{I_E = I_B + I_C}$ (saturation / active)

NPN

- $V_{BE} = V_B - V_E$
  (red) (Black)
- $V_{CE} = V_C - V_E$
- $V_{CB} = V_C - V_B$

PNP

- $V_{EB} = V_E - V_B$
  (red) (Black)
- $V_{BC} = V_B - V_C$
- $V_{EC} = V_E - V_C$

Parameters (mainly for forward active mode)

1. $\beta = \dfrac{I_C}{I_B}$, $\beta \approx 50-250$
2. $\alpha = \dfrac{I_C}{I_E}$, $\alpha \approx 1$

Scanned with CamScanner

Equations :

• $I_c = \beta I_B$ • $I_c = \alpha I_E$

★ $I_E = I_B + I_c$ (applicable for all modes)

$I_E = I_B + \beta I_B$

★ $I_E = I_B(1+\beta)$

$I_E = I_B + I_c$

• $I_B = \frac{I_c}{\beta}$ $I_E = \frac{I_c}{\beta} + I_c$

★ $I_E = I_c\left(\frac{1}{\beta} + 1\right)$

$$\frac{I_E}{I_c} = \frac{1}{\alpha}$$

$$\frac{I_E}{\beta I_B} = \frac{1}{\alpha}$$

$$\frac{1}{\alpha} = \frac{\cancel{I_B}(1+\beta)}{\beta \cancel{I_B}}$$

★ $\alpha = \frac{\beta}{1+\beta}$ $\frac{I_c}{I_E} = \frac{1}{\alpha} = \frac{1+\beta}{\beta}$

- $I_E = I_B + I_C$

$$\frac{I_C}{\alpha} = \frac{I_C}{\beta} + I_C$$

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\frac{1}{\beta} = \frac{1}{\alpha} - 1$$

$$\star \quad \beta = \frac{\alpha}{1-\alpha}$$

- Common Emitter Configuration (iskay ref say measurement)

  Base → Input terminal − I_B, V_{BE}
  Collector → Output terminal − I_C, V_{CE}

- Common Base Configuration

  Emitter → Input terminal − I_E, V_{BE}
  Collector → Output terminal − I_C, V_{CB}

- Common Collector Configuration

  Base → Input terminal − I_B, V_{CB}
  Emitter → Output terminal − I_C, V_{CE}

- Base can never be output terminal (always input)
- Collector can never be input terminal (always output).

- Voltages taken w.r.t NPN/PNP

Scanned with CamScanner

Monday 10th Oct 2022

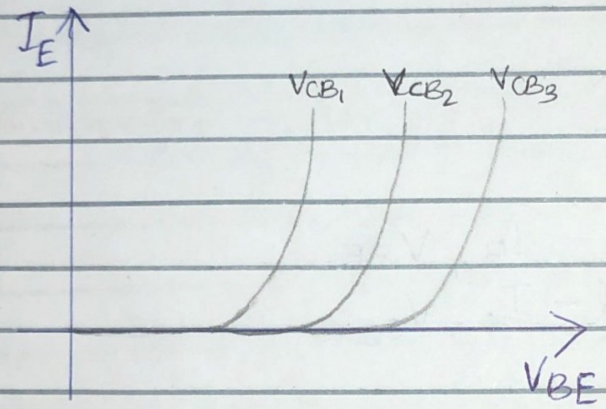
Input Characteristics:

A graph between input voltage & input current for constant output voltage

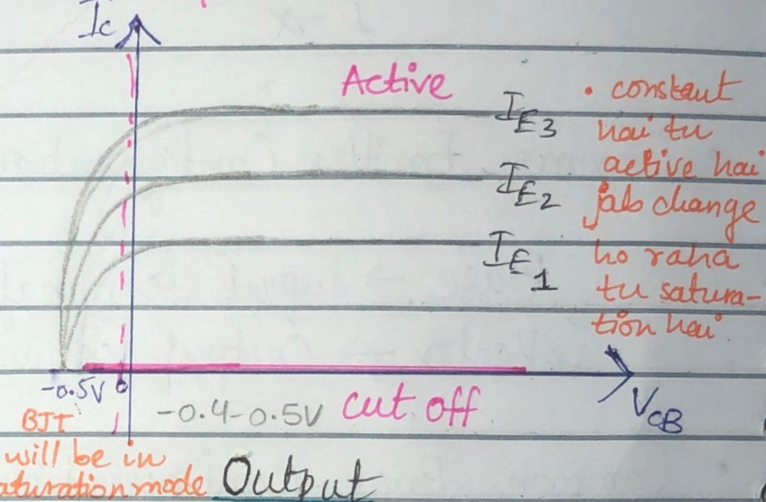
Output Characteristics:

A graph between output voltage & output current for constant input current.

① Common Base (CB)



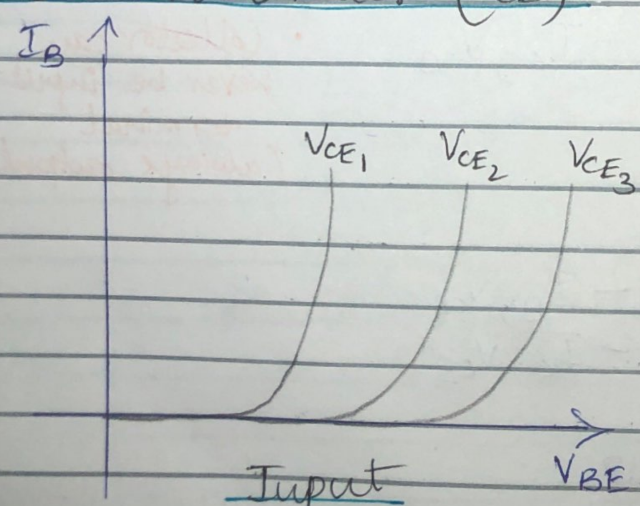
Input



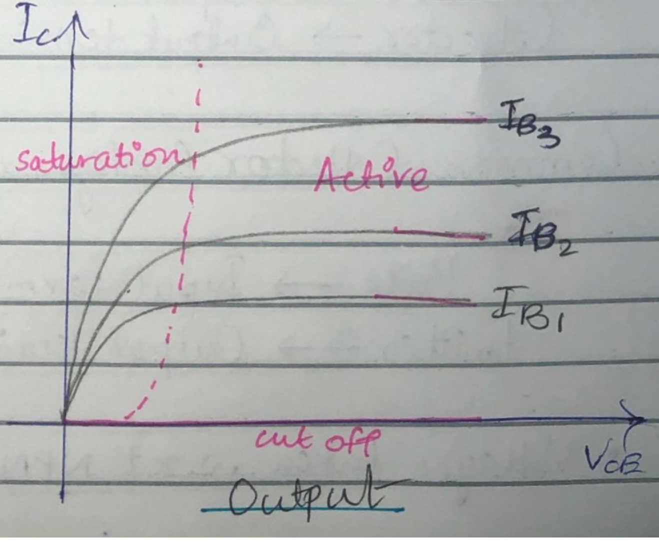
Output

② Common Emitter (CE)

$I_E = (\beta + 1) I_B$ (Forward Active)

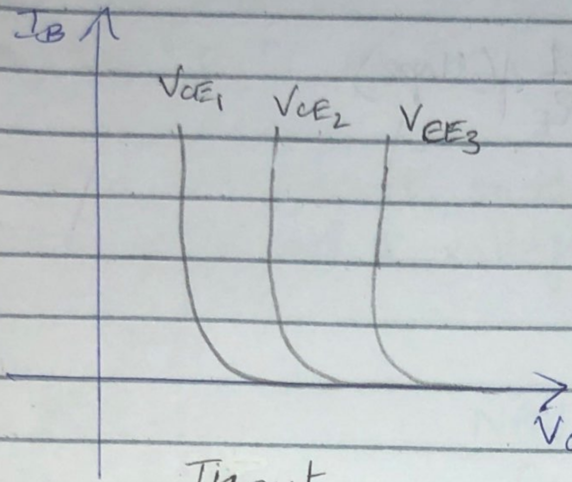


Input

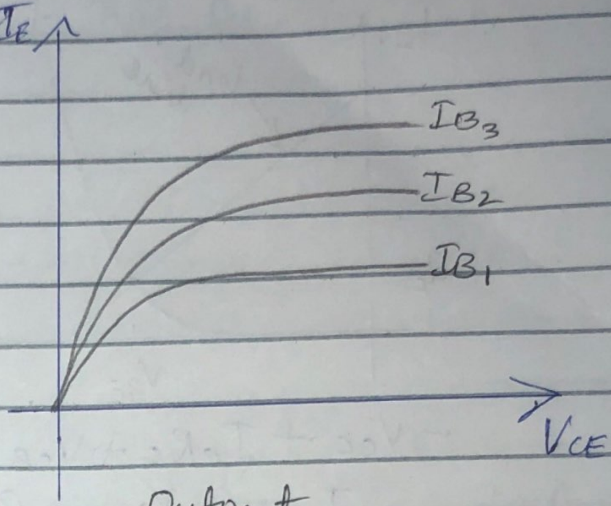


Output

③ Common Collector



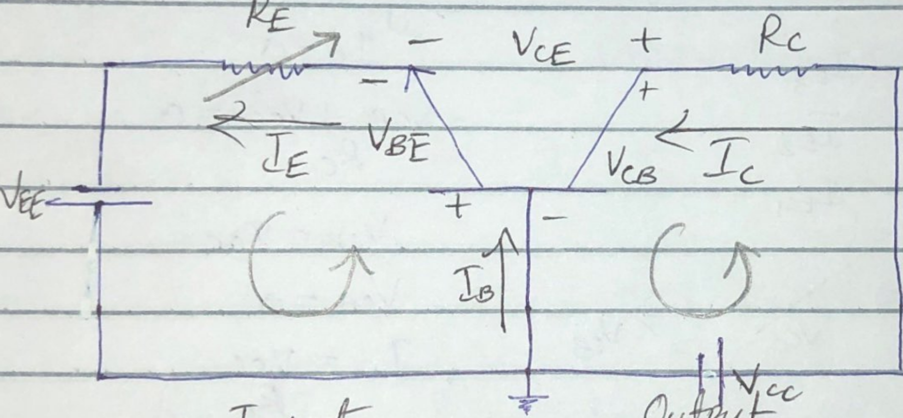
Input



Output

$I_C < I_E$

- BJT acts as a constant current source in active mode.



NPN

Common Base Configuration.

Input: I_E, V_{BE}

Output: I_C, V_{CB}.

Input: $-V_{EE} + V_{BE} + I_B R_E = 0$.

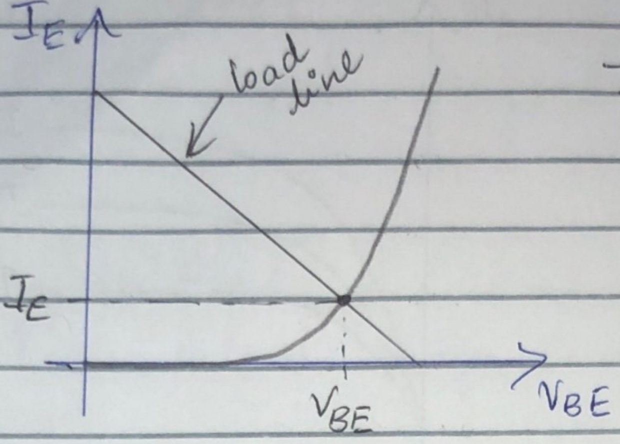
Output: $-V_{CC} + V_{CB} + I_C R_C = 0$.

$$I_E = \frac{-V_{BE}}{RE} + \frac{V_{EE}}{RE}$$

$$y = mx + c$$

$$m \downarrow -\frac{1}{R_E} \qquad c \downarrow \frac{V_{EE}}{R_E}$$

Graphical Analysis



$\frac{-1}{R_E}$ (slope)

$$-V_{cc} + I_c R_c + V_{cB} = 0$$

$$I_c = \frac{-V_{cB}}{R_c} + \frac{R_{ce}}{R_c}$$

$y = mx + c$

$I_c = 0$

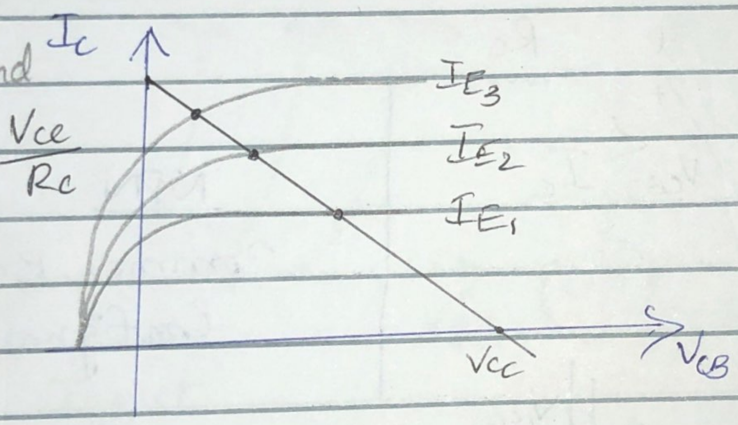
$\frac{-V_{cB} + V_{cc}}{R_c} = 0$

$V_{cB} = V_{cc}$

$V_{cB} = 0$

$I_c = \frac{V_{cc}}{R_c}$

- Output and Input ka same point lein gey



Mathematical Analysis

Input

Assume $V_{BE} = 0.7V$

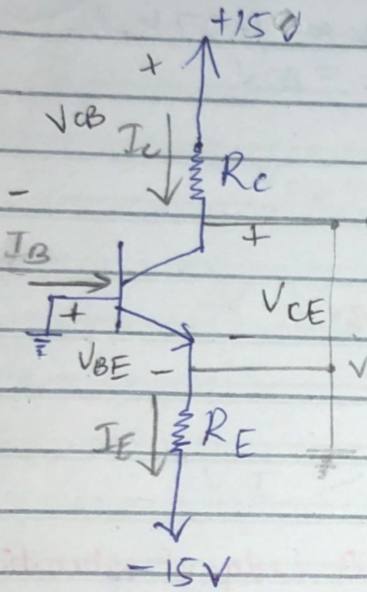
Output

we need α and β

Wednesday 12th Oct 2022

Q: The transistor in the circuit has $\beta = 100$ and exhibits a $V_{BE} = 0.7V$ at $I_C = 1mA$

Design the circuit so that a current of 2mA flows through the collector and a voltage of +5V appears at collector.



NPN

Common Base

Input : I_E, V_{BE}

Output : I_C, V_{CB}

$I_C = 2mA$

$V_C = 5V$

$V_B = 0V$

$I_E = \frac{I_C}{\alpha}$ $\qquad$ $\alpha = \frac{\beta}{1+\beta} = \frac{100}{101} = 0.99$

$I_E = \frac{2m}{0.99} = 2.02mA.$

$R_C = ?$ $\qquad R_C = \frac{V_{CC} - V_C}{I_C} = \frac{15-5}{2m} = 5k\Omega$

$R_E = ?$ $\qquad R_E = \frac{V_E - (-15)}{I_E}$

★ $\boxed{I_C = I_S e^{V_{BE}/V_T}}$ $\qquad n=1$ always in BJT

$V_{BE_2} = V_{BE_1} + 2.3 n V_T \log \frac{I_2}{I_1}$

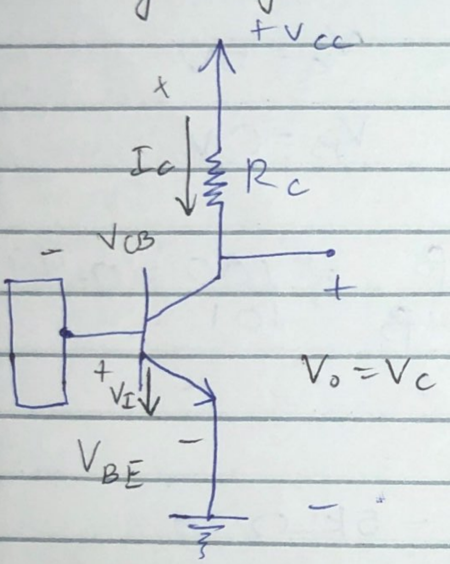
$V_{BE} = 0.7 + 2.3(1)(25m) \log \frac{2m}{1m} = 0.717V$

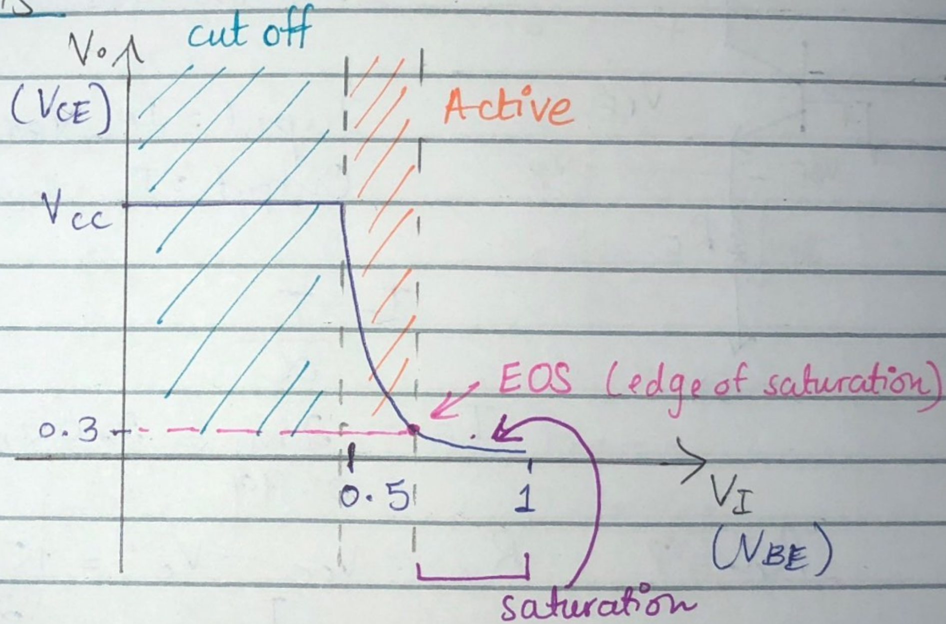
$$V_{BE} = V_B - V_E$$

$$0.717 = 0 - V_E$$
$$V_E = -0.717V$$

$$R_E = \frac{-0.717 + 15}{2.02m} = 7070.80\Omega \approx 7.07k\Omega$$

Large Signal Analysis





$$V_c = V_{CE} = V_o = V_{cc} - i_c R_c$$

① $V_I < 0.5V$ (it starts conduction according to our assumption)

BEJ is reverse biased

cut off $i_B = 0 \Rightarrow i_c$

② $V_I > 0.5V$

$I_C = \beta I_B$ $\quad I_C = I_S e^{V_{BE}/V_T}$

V_{BE} = forward biased.

The transistor is going from cutoff to Active mode.

V_{CB} = $-ve$ = forward biased.

$V_{CB} = V_C - V_B$ (If base will be at higher potential P higher than N)

☆ $V_{CB} < -0.4$ (saturation)

active mode

$V_I > 0.5V$

$$V_O = V_{CC} - \overbrace{I_S e^{V_{BE}/V_T}}^{I_C} R_C$$

$$V_O = V_{CC} - I_C R_C$$

$V_{BC} > 0.4V$

$V_{CB} = -0.4V$ (edge of saturation).

is the last point of active mode after which transistor enters saturation mode.

☆ $V_{CB} = V_{CE} - V_{BE}$

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$$V_{CB} = V_{CE} - V_{BE}$$

$$= V_C - \cancel{V_E} - [V_B - \cancel{V_E}]$$

$$= V_C - V_B$$

$$-0.4 = V_{CE} - 0.7$$

$$\boxed{V_{CE} = 0.3V} \text{ (EOS)}$$

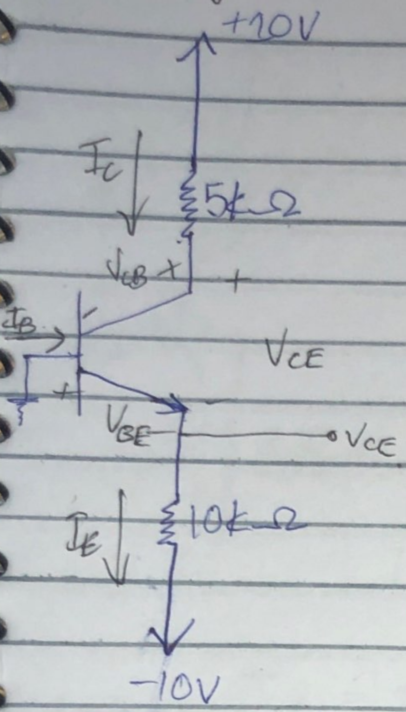
Edge of Saturation:

$$V_{CE(EOS)} = 0.3 \quad , \quad I_C(EOS)$$

For Saturation:

$$V_{CE(SAT)} = 0.2V \text{ (less than 0.3)} , \quad I_C(sat).$$

Monday 17th Oct 2022



→ $\beta = 50$

→ Analyze the circuit.

find $I_C, I_B, I_E = ?$

$V_C, V_B, V_E = ?$

Operating Mode = ?

↳ Assume Active mode

$V_{BE} = 0.7V$

$V_{CB} = V_C - V_B$

$I_E = I_B + I_C$

$V_{BE} = V_B - V_E$

$0.7 = 0 - V_E$

$V_E = -0.7$

$$I_E = \frac{-0.7 - (-10)}{10 \times 10^3} = 9.3 \times 10^{-4}$$

$$\alpha = \frac{\beta}{1+\beta} = \frac{50}{51} = 0.9804$$

$$I_C = I_E \alpha = (9.3 \times 10^{-4})(0.9804)$$
$$= 9.118 \times 10^{-4} = 0.912 mA$$

$$I_B = \frac{I_C}{\beta} = \frac{0.912m}{50} = 18.24 \mu A$$

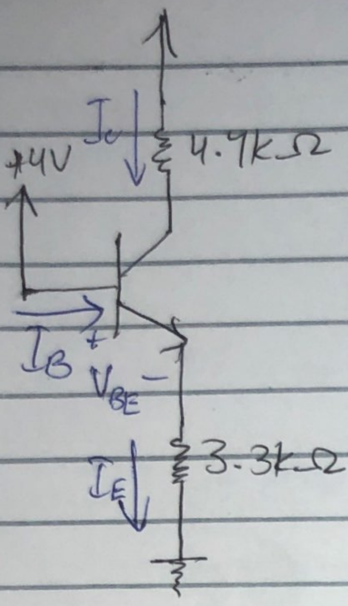
$V_C = 10 - I_C R_C$

$V_C = 10 - (0.912 \times 10^{-3})(5 \times 10^3)$

$= 5.44V$

$V_{CB} = V_C - V_B = 5.44 - 0 = 5.44$.

(not less than −0.4V, so assumption is correct and BJT is operating in active mode).



$\beta = 100$

$\alpha = \frac{100}{1+100} = 0.99$

Assume active mode

$V_{BE} = 0.7V$

$V_{BE} = V_B - V_E$

$V_B = 4$

$0.7 = 4 - V_E$

$V_E = 3.3$.

$I_E = \frac{3.3}{3.3 \times 10^3} = 1\,mA$

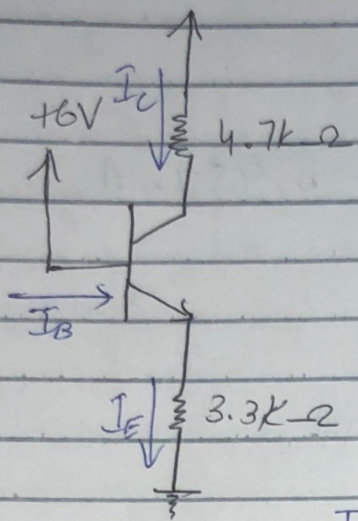
$I_c = \alpha I_E = 0.99 \times 1 \times 10^{-3} = 990\,\mu A$

$I_B = \frac{I_c}{\beta} = \frac{990 \times 10^{-6}}{100} = 9.9\,\mu A$

$V_c = 10 - I_c R_c = 10 - (990\mu)(4.7k) = 5.36V$

$V_{CB} = V_c - V_B = 5.36 - 4 > -0.4V$

$= 1.36 > -0.4V$



$\beta = 100$

$\alpha = 0.99$

$V_{BE} = 0.7$

$V_{BE} - V_B = -V_E$

$0.7 - 6 = -V_E$

$V_E = 5.3$

$I_E = \frac{V_E - 0}{R_E} = \frac{-5.3}{3.3k} = 1.61\,mA$

$I_c = 1.59\,mA$

$I_B = 15.9\,\mu A$

$V_C = 2.53V$

$V_{CB} = 2.53 - 6 < -0.4$

$= -3.47 < 0.4$

(saturation mode)

So, Beta won't be constant.

For Saturation Mode:

$\beta_{forced} = \beta_f = \frac{I_c\,(sat)}{I_B}$

$V_{CE}\,(sat) = 0.2V$

$V_{CE}\,(sat) = V_C\,(sat) - V_E$

$0.2 = V_c\,(sat) - 5.3$

$V_c(sat) = 5.5V$

$$I_c = \frac{10 - V_c(sat)}{R_c} = \frac{10-5.5}{4.7k} = 0.957mA$$

$$I_B = I_E - I_C$$

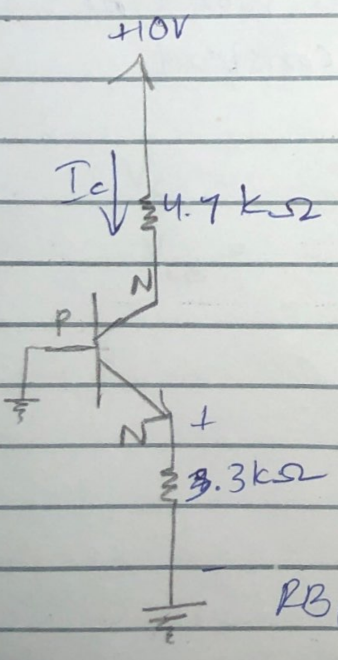
$$= 1.61 - 0.957 = 0.653mA$$

$$V_{CB} = V_C - V_B$$
$$= 5.5 - 6$$
$$= -0.5 < -0.4V$$

BJT is in Saturation mode.

$$I_c = \beta I_B$$

$$\beta_f = \frac{I_c}{I_B} = \frac{0.957}{0.653} = 1.49.$$



BEJ RB

that means $I_B = I_C = I_E = 0$

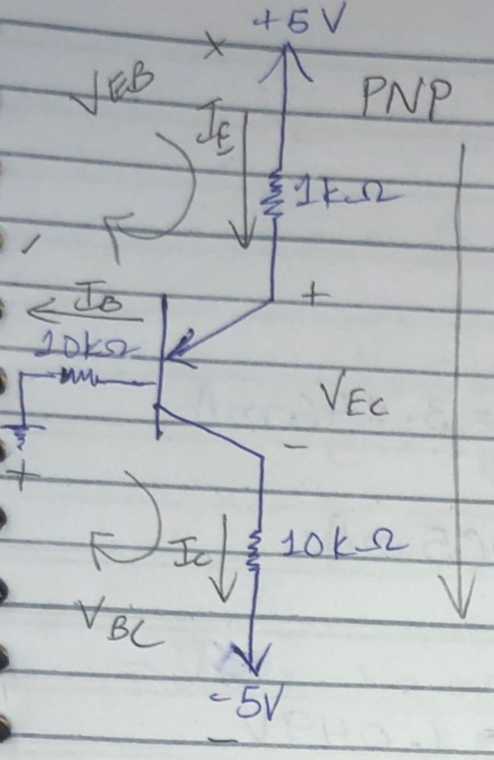
$$V_E = V_B = 0$$

$$V_{CC} = V_C = 10$$

RB (Collector Base Junction)

19th Oct 2022

Wednesday



$\beta = 30$

Analyze the circuit

Assume active mode

$V_{EB} = 0.7$

$V_B - V_C < -0.4V$

$\alpha = \frac{\beta}{1+\beta} = \frac{30}{31} = 0.968$

$$I_E = I_B + I_C$$

$$V_{EB} = V_E - V_B$$

$$V_B = I_B R_B = (10k)(I_B)$$

$$-5 + I_E R_E + V_{EB} + I_B R_B = 0$$

$$I_E R_E + I_B R_B = 4.3$$

$$I_E = (\beta + 1) I_B$$

$$I_B = \frac{I_E}{\beta + 1}$$

$$I_E R_E + \left(\frac{I_E}{31}\right) R_B = 4.3$$

$$I_E \left(R_E + \frac{R_B}{31}\right) = 4.3$$

Scanned with CamScanner

$$I_E = \frac{4.3}{(1k) + \frac{(10k)}{31}} = 3.25mA$$

$$I_c = \alpha I_E$$

$$I_c = (0.968)(3.25 \times 10^{-3}) = 3.146mA$$

$$I_B = \frac{I_c}{\beta} = \frac{3.146m}{30} = 0.105mA$$

$$V_B = I_B R_B = (0.105)(10k) = 1.049V$$

$$V_E = 1.04 + 0.7 = 1.74V$$

$$V_c = I_c R_c - 5 = (3.14m)(10k) - 5$$
$$= 31.4 - 5 = 26.4V$$

BJT is in saturation mode

$$V_B - V_c = 1.04 - 26.4 = -25.36$$

$$I_E = I_B + I_c$$

$$V_{EB} = 0.7V = V_E - V_B$$

$$V_{EC}(sat) = 0.2V = V_E - V_c$$

$$I_E = \frac{5 - V_E}{R_E} \qquad I_c = \frac{V_c - (-5)}{R_c}$$

$V_E = V_B + 0.7$

$I_E = \frac{5 - (V_B + 0.7)}{R_B} = \frac{4.3 - V_B}{R_E}$

$V_C = V_E - 0.2$

$V_C = V_B + 0.7 - 0.2$

$V_C = V_B + 0.5$

$I_C = \frac{V_B + 0.5 + 5}{R_C} = \frac{V_B + 5.5}{R_C}$

$\frac{4.3 - V_B}{R_E} = \frac{V_B}{R_B} + \frac{V_B + 5.5}{R_C}$

$\frac{4.3 - V_B}{1k} = \frac{V_B}{10k} + \frac{V_B + 5.5}{10k}$

$10k(4.3 - V_B) = 1k(2V_B + 5.5)$

$43k - 10kV_B = 2k V_B + 5.5k$

$37.5k = 12k V_B$

$V_B = 3.125$

$V_C = 3.625V$

$V_E = 3.825V$

$I_B = 0.313 mA$

$I_C = 0.863 mA$

$$\beta_f = \frac{I_C}{I_B} = \frac{0.863}{0.312} = 2.8$$

$I_E = 1.175mA$

$V_B - V_C = 3.125 - 3.625 = -0.5$

$V_B - V_C < -0.4V$

So, BJT is in saturation mode.

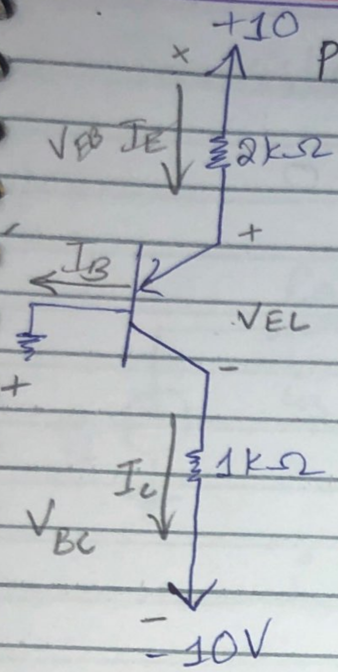
★ Overdrive factor

How deep BJT is in saturation mode.

$$\frac{\beta}{\beta_f} = \frac{I_B}{I_B(EOS)} = \frac{30}{2.8} = 10.7$$

$$\frac{0.104m}{0.312m} = 0.333$$

• Min value of overdrive factor < 1



PNP

$\beta = 100$ $\alpha = 0.99$

Active Mode

$V_{EB} = 0.7$

$V_{EB} = V_E - V_B$

$0.7 = V_E - 0$

$V_E = 0.7$

$I_E = \dfrac{10 - 0.7}{2 \times 10^3} = 4.65mA$

$I_C = \alpha I_E = 0.99 \times 4.65m = 4.604mA$

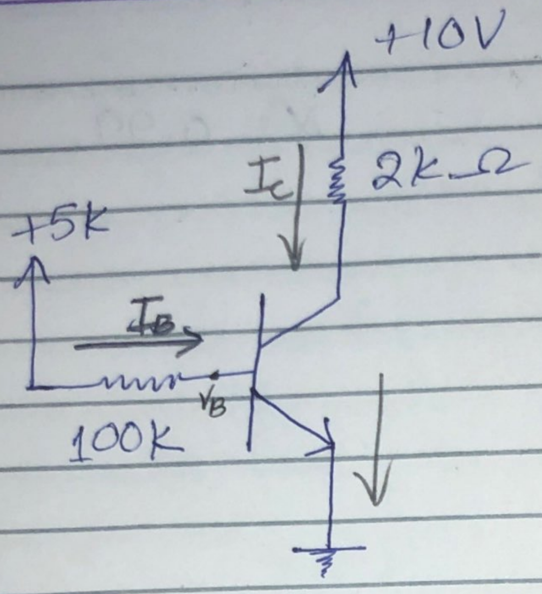
✓ $I_E = I_B + I_C$ or $I_B = \dfrac{I_C}{\beta} = 46\mu A$

$V_c = -5.4V$

$V_B - V_c$

$0 - (-5.4) > -0.4V$

So, BJT in active mode.



$\beta = 100$

$V_{BE} = 0.7V \quad V_E = 0$

$V_{BE} = V_B - V_E$

$0.7 = V_B - 0.$

$V_B = 0.7V$

$-5 + I_B R_B + V_{BE} = 0.$

$$I_B = \frac{5 - 0.7}{100k} = \frac{4.3m}{100} = 43\mu A.$$

$I_c = \beta I_B = 4.3mA$

$V_c = V_{cc} - I_c R_c = 10 - (4.3)(2) = 1.4V$

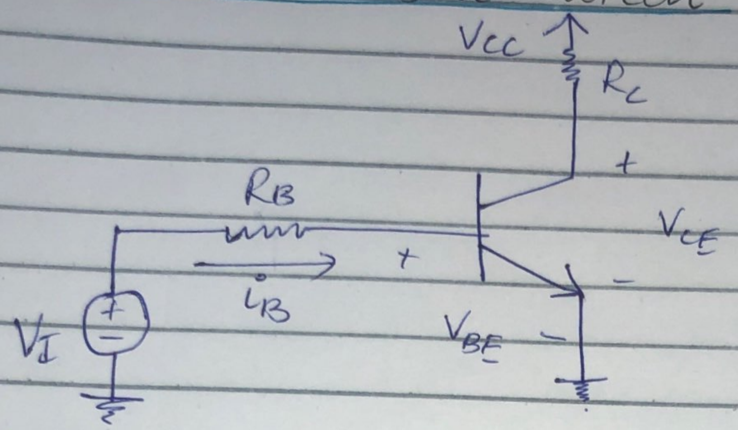
$V_{CB} = V_c - V_B = 1.4 - 0.7 \geq 0.4$

So, BJT is in active mode.

β dependent design..

β is temperature dependant. So, this circuit is highly dependent on temperature as it is highly dependent on β.

Transistor as a Switch



$V_{BE} = 0.7V$

$V_I > 0.5V$

$$i_B = \frac{V_I - V_{BE}}{R_B} \qquad i_c = \beta i_B$$

$$I_{c(EOS)} = \frac{V_{cc} - V_{CE(EOS)}}{R_c} = \frac{V_{cc} - 0.3}{R_c}$$

$$I_c(EOS) = \beta I_B(EOS)$$

$$V_I(EOS) = I_B(EOS) R_B + V_{BE}$$

$$\beta_f = \frac{I_c(sat)}{I_B}$$

$$\text{Overdrive} = \frac{\beta}{\beta_f} = \frac{I_B}{I_B(EOS)}$$

$$V_{CE}(sat) = 0.2V$$

$$I_c(sat) = \frac{V_{cc} - 0.2}{R_c}$$